

1. What is the geometrical structure of the Puma 560 robot?

*Serial chain manipulator with 6 revolute joints*

2. Fill the DH parameter table for Puma 560 robot.

| $j$ | $\theta$ | $d$     | $a$    | $\alpha$ |
|-----|----------|---------|--------|----------|
| 1   | $q_1$    | 0       | 0      | 1.5708   |
| 2   | $q_2$    | 0       | 0.4318 | 0        |
| 3   | $q_3$    | 0.15005 | 0.0203 | -1.5708  |
| 4   | $q_4$    | 0.4318  | 0      | 1.5708   |
| 5   | $q_5$    | 0       | 0      | -1.5708  |
| 6   | $q_6$    | 0       | 0      | 0        |

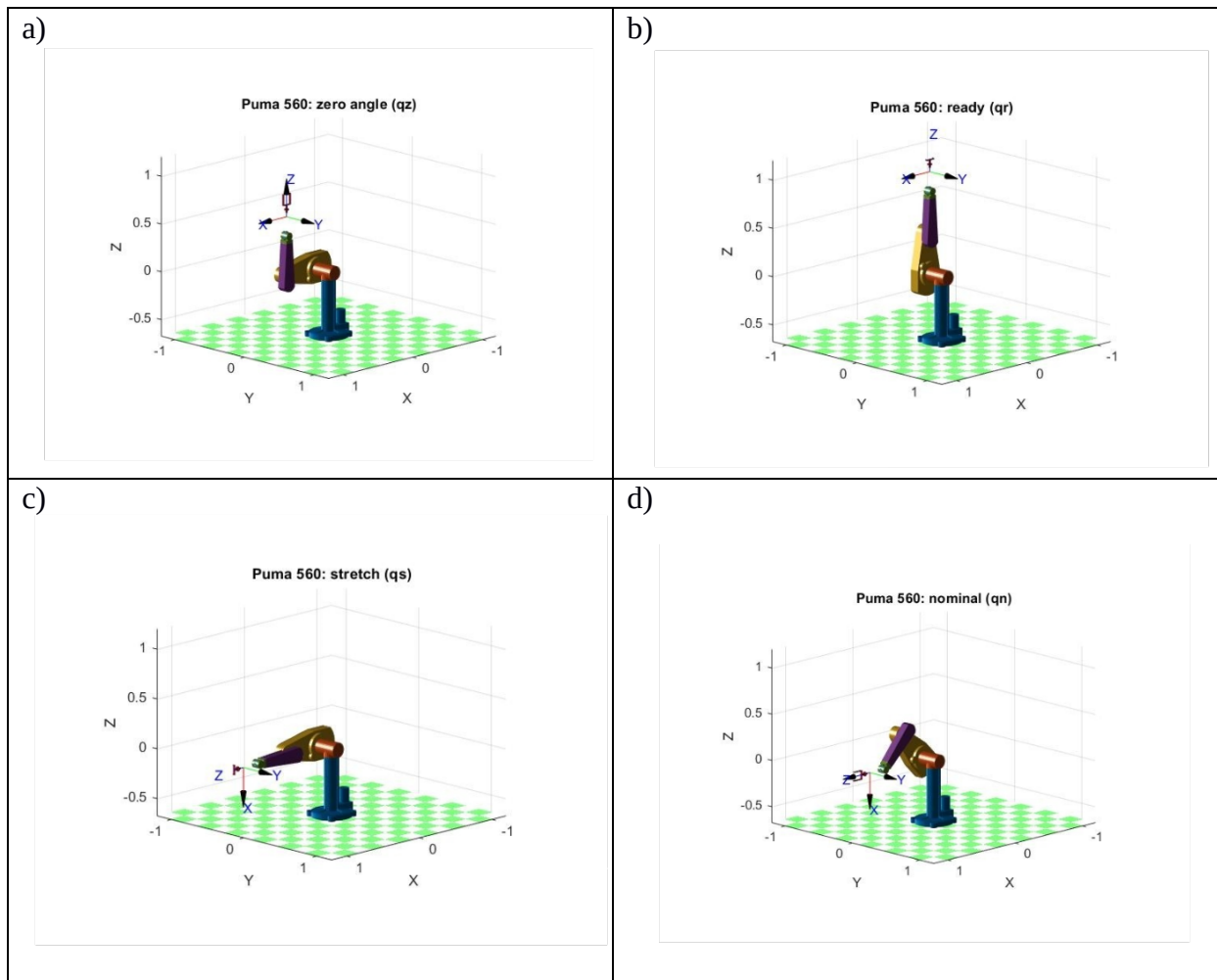
3. Joint coordinate vectors of Puma 560 robot for the following canonical configurations:

- a) zero angle:  $[0 \ 0 \ 0 \ 0 \ 0 \ 0]$   
 b) ready :  $[0 \ 1.5708 \ -1.5708 \ 0 \ 0 \ 0]$   
 c) stretch :  $[0 \ 0 \ -1.5708 \ 0 \ 0 \ 0]$   
 d) nominal :  $[0 \ 0.7854 \ 3.1416 \ 0 \ 0.7854 \ 0]$

4. Forward kinematics for tool center point (TCP) in Procedure 3.5 for the canonical configurations.

| Configuration | Position                                                     | Orientation (Rotation Matrix)                                        |
|---------------|--------------------------------------------------------------|----------------------------------------------------------------------|
| Zero angle    | $\begin{bmatrix} 0.4521 \\ -0.1500 \\ 0.6318 \end{bmatrix}$  | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  |
| Ready         | $\begin{bmatrix} 0.0203 \\ -0.1500 \\ 1.0636 \end{bmatrix}$  | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  |
| Stretch       | $\begin{bmatrix} 1.0636 \\ -0.1501 \\ -0.0203 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$ |
| Nominal       | $\begin{bmatrix} 0.7963 \\ -0.1501 \\ -0.0144 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$ |

5. Visualization of Puma 560 robot for Procedure 3.6.

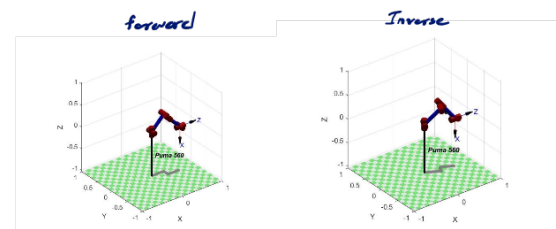


6. Inverse kinematics joint vector for Procedure 3.8. What is your observation?

The resulting  $q_{inv}$  matches the original  $q_n$  with having minor deviations. This confirms that the  $FK \rightarrow IK$  mapping is consistent & the solver is working correctly for reachable configurations.

7. Correct arm configuration for Procedure 3.9.

c) re- Right hand elbow up



8. What can be observed for Procedure 3.10?

```
Warning: point not reachable
> In SerialLink/ikine6s (line 573)
In roboticslab3 (line 73)
NaN NaN NaN NaN NaN NaN
Unreachable: ikine6s returned NaNs (no solution).
```

## 9. MATLAB code for the entire procedure.

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| <pre> 1 %% Tasks for Lab 03 (Puma 560) 2 %% 3.1 Instantiate Puma 560 robot model 3 mdl_puma560 % built-in model from the Robotics Toolbox 4 fprintf('\n Puma 560 object created \n'); 5 6 %% 3.2 - Robot geometry and DH parameter table 7 p560: %Puma 560 (Unimation):: 6 axis, RRRRRR, stdDH, slowRNE - viscous friction; params of 8/95; 8 9 %% 3.3 - Joint coordinate vectors for canonical configurations 10 % Canonical Configurations (Joint coordinate Vectors) 11 qz, qr, qs, qn 12 13 %% 3.4 - Define tool transform: extension of 200 mm in z of T6 frame 14 p560.tool = SE3(0,0,0.2); % extend 200mm along z-axis of T6 15 16 %% 3.5 Forward kinematics for canonical configurations 17 configs = ['zero angle (qz)', 'ready (qr)', 'stretch (qs)', 'nominal (qn)']; 18 q_vecs = {qz,qr,qs,qn}; 19 20 for i = 1:4 21     q = q_vecs{i}; 22     T = p560.fkine(q); 23     fprintf('\nConfiguration: %s\n', configs{i}); 24     fprintf('Position (m): '); disp(transl(T)); 25     disp('Rotation matrix:'); disp(rotz(T)); 26 end 27 28 %% 3.6 Graphically display realistic plots for canonical configurations 29 for i = 1:4 30     figure('Name',sprintf('Puma560 - %s',configs{i})); 31     % plot3d provides more realistic model (shaded links) if supported 32     p560.plot3d(q_vecs{i}) 33     title(sprintf('Puma 560: %s', configs{i})); 34 end 35 36 %% 3.7 reset the tool transform to zero extension 37 p560.tool = SE3(0,0,0); % extend 200mm along z-axis of T6 38 39 %% 3.8 perform inverse kinematics 40 T_nominal = p560.fkine(qn); 41 disp('Target T_nominal:'); disp(T_nominal.T); 42 %p560.plot(qn) 43 44 % Analytical inverse kinematics (for 6-axis spherical wrist) 45 q_inv_nominal = p560.ikine6s(T_nominal); 46 disp('Recovered joint vector (ikine6s):'); disp(q_inv_nominal); 47 %p560.plot(q_inv_nominal) 48 49 fprintf('Difference from qn (radians):\n'); 50 disp(q_inv_nominal - qn); 51 52 T_check = p560.fkine(q_inv_nominal); 53 disp('FK result from IK solution:'); disp(T_check.T); 54 fprintf('Pose difference (should be near zero):\n'); 55 disp(T_check.T - T_nominal.T); 56 </pre> | <pre> 57 %% 3.9 Inverse Kinematics for a New Pose 58 configs = {'lu','ld','ru','rd'}; % 3.9 options (wrist default = 'n') 59 sol = struct([]); 60 61 for i = 1:numel(configs) 62     q_try = p560.ikine6s(T_nominal, configs{i}); 63     sol(i).cfg = configs{i}; 64     sol(i).q = q_try; 65     sse = sum((q_try - qn).^2); % how close to the provided nominal vector 66     sol(i).sse = sse; 67     fprintf('%s SSE: %.4f\n', configs{i}, sse); 68     figure: p560.plot3d(q_try); title(['IK: ', configs{i}]); 69 end 70 71 %% 3.10 Give an unreachable point 72 T_bad = SE3(2.0, 0, 0); % way beyond reach 73 q_bad = p560.ikine6s(T_bad); % spherical-wrist IK 74 disp(q_bad) 75 if any(isnan(q_bad)) 76     disp('Unreachable: ikine6s returned NaNs (no solution).'); 77 end 78 79 %% observe Puma 560 motion 80 close all; clear all; 81 mdl_puma560 82 T1 = SE3(0.8, 0, 0) * SE3.Ry(pi/2); 83 T2 = SE3(-0.8,0,0) * SE3.Rx(pi); 84 85 % IK (spherical wrist) 86 q1 = p560.ikine6s(T1); 87 q2 = p560.ikine6s(T2); 88 89 % Quintic joint trajectory 90 t = (0:0.05:2)'; % time vector 91 q = traj(q1, q2, t); % joint waypoints 92 93 % Animate realistic 3D model 94 p560.plot3d(q); 95 96 </pre> |
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## 10. Explain in point form what the MATLAB code in 3.11 does.

- `mdl_puma560` - loads the predefined 6-DOF PUMA 560 Model.
- `T1, T2` - create 2 target tool poses:
  - `T1`: 0.8 m along  $+x$ , oriented  $90^\circ$  about  $Y$ .
  - `T2`: 0.8 m along  $-x$ , oriented  $180^\circ$  about  $x$ .
- `ikine6s` - compute inverse kinematics joint angles for each target pose.
- `traj(q1, q2, t)` - generate a smooth joint-space trajectory between  $q_1$  &  $q_2$ , using a quintic polynomial with zero initial & final velocity.
- `plot3d(q)` - animates a realistic 3-D motion of the puma 560 executing that trajectory.

Observation: Robot moves smoothly from 1<sup>st</sup> pose to 2<sup>nd</sup> pose through physically valid intermediate configurations. It demonstrates forward & inverse kinematics & trajectory generation working together.